Qihua Dong (He/Him/His)

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EDUCATION

Northeastern University, Boston, USA Sep 2023 – Dec 2027 (expected)
PhD in Computer Engineering; GPA: 3.96/4.00

City University of Hong Kong, Hong Kong SAR, China Sep 2017 – Jul 2021
BSc in Computer Science with Mathematics Minor; GPA: 3.89/4.3 (Top 10%)

RESEARCH INTERESTS

Reasoning, post-training, and self-improving agents in (M)LLMs, including reinforcement learning, tool use, and agent harnesses.

SELECTED PUBLICATIONS

Publications

**: Equal contribution.*

- **Qihua Dong**, Ruozhen He, Yizhou Wang, Jianglin Lu, Yiyang Huang, Xu Ma, Hailing Wang, Su Yuan, Huimin Zeng, Mingyuan Zhang, Liang Shi, Yun Fu. "RetryScribe: Retry-Aware Captioning for Robotic Manipulation Videos." *Under submission*, 2026.
- **Qihua Dong**, Ruozhen He, Junwen Chen, Yizhou Wang, Xu Ma, Songyao Jiang, Yun Fu. "Hierarchical Visual Agent: Managing Contexts in Joint Image-Text Space for Advanced Chart Reasoning." *ACL Findings* 2026. arXiv:2605.04304
- **Qihua Dong**, Yitian Zhang, Huimin Zeng, Yizhou Wang, Jianglin Lu, Kuo Yang, Yun Fu. "Revealing the Seen, Imagining the Beyond: A Survey of Image-Grounded Chain-of-Thought Reasoning in Multimodal LLMs." *ACL Main* 2026. ACL Anthology
- **Qihua Dong**, Gozde Sahin, Pei Wang, Zhaowei Cai, Robik Shrestha, Hao Yang, Davide Modolo. "Visual Reasoning through Tool-supervised Reinforcement Learning." *CVPR Findings* 2026. arXiv:2604.19945
- **Qihua Dong**, Yang Kuo, Ju Lin, Handong Zhao, Yitian Zhang, Yizhou Wang, Huimin Zeng, Jianglin Lu, Yun Fu. "Ref-Adv: Exploring MLLM Visual Reasoning in Referring Expression Tasks." *ICLR* 2026. arXiv:2602.23898
- Xu Ma, Yitian Zhang, **Qihua Dong**, Yun Fu. "Fine-T2I: An Open, Large-Scale, and Diverse Dataset for High-Quality T2I Fine-Tuning." *arXiv preprint*, 2026. arXiv:2602.09439
- **Qihua Dong**, Luis Figueroa, Handong Zhao, Kushal Kafle, Jason Kuen, Zhihong Ding, Scott Cohen, Yun Fu. "CoT Referring: Chain-of-Thought Grounding Improve Localization in Referring Expression Tasks." *arXiv preprint*, 2025. arXiv:2510.06243
- **Qihua Dong**, Hao Du, Ying Song, Yan Xu, Jing Liao. "Preserving Tumor Volumes for Unsupervised Medical Image Registration." *ICCV* 2023. arXiv:2309.10153
- Ruozhen He*, **Qihua Dong***, Jiayin Lin, Rynson Lau. "Weakly-Supervised Camouflaged Object Detection with Scribble Annotations." *AAAI* 2023. arXiv:2207.14083
- Hao Du*, **Qihua Dong***, Yan Xu, Jing Liao. "Weakly-Supervised 3D Medical Image Segmentation using Geometric Prior and Contrastive Similarity." *IEEE TMI* 2023. arXiv:2302.02125

Academic Service

- **Outstanding Reviewer**, ICCV 2025.

INDUSTRY EXPERIENCE

Meta, New York, NY, USA May 2026 – Aug 2026
Research Scientist Intern
Mentor: Daniel Bolya
Worked on multimodal LLM post-training and reasoning, including midtraining, RL, and evaluation.

Amazon, AGI Foundations Team, WA, USA May 2025 – Nov 2025
Applied Scientist Intern
Mentors: Zhaowei Cai, Pei Wang, Hao Yang, Gozde Sahin, Robik Shrestha, Davide Modolo
Built tool-use visual agents (zoom, draw line/point, rotate/flip) with multi-turn GRPO-based training. Applied curriculum RL and created data pipelines for 7B/32B models with novel reward design, improving performance across chart, document, and high-resolution image benchmarks.

Adobe Research, CA, USA May 2024 – Feb 2025
2024 Intern - Research Scientist/Engineer
Mentors: Luis Figueroa, Handong Zhao, Kushal Kafle, Jason Kuen, Zhihong Ding, Scott Cohen
Developed multimodal LLM for referring image segmentation with structured chain-of-thought reasoning. Created novel benchmarks with enriched annotations for complex referring expressions.

RESEARCH EXPERIENCE

Stipended Graduate Research Assistant (PhD Student) Sep 2023 – Present
Northeastern University
Building multimodal LLMs capable of visual grounding, spatial reasoning, and zero-shot generalization for intelligent agent development with minimal fine-tuning via in-context examples.

Research Assistant May 2022 – Mar 2023
City University of Hong Kong (Advisors: Prof. Jing Liao, Prof. Yan Xu)
Conducted research on 3D medical imaging. Developed a volume-preserving registration network for tumor alignment with a task-specific loss. Designed a dynamic transformer for efficient medical image segmentation via foreground-aware tokenization. Proposed a weakly supervised segmentation framework combining geometric priors and contrastive pretraining, achieving 12.5% Dice improvement on benchmark datasets.

Research Assistant Dec 2021 – Jun 2022
City University of Hong Kong (Advisor: Prof. Rynson Lau)
Conducted research on weakly-supervised camouflaged object detection. Initiated the first weakly supervised formulation and dataset for camouflaged object detection with new consistency and feature losses, surpassing SOTA COD baselines by 11% MAE.

ADDITIONAL EXPERIENCE

Texas A&M University, TX, USA Oct 2022 – Jan 2023
Research Assistant (Remote) with Prof. Wenping Wang; contributed to computer vision and computer graphics projects focused on facial analysis.

City University of Hong Kong, Hong Kong SAR, China Sep 2020 – Feb 2021
Part-time Student Research Assistant with Prof. Jianping Wang; researched scheduling schemes for self-driving systems.

HONORS AND SCHOLARSHIPS

- First Class Honors, City University of Hong Kong, 2021.
- Bronze Medal, ICPC Asia Nanchang Regional Contest, 2019.
- Dean's List, City University of Hong Kong, 2017–2021.
- Full Tuition Scholarship, City University of Hong Kong, 2017–2021.

SKILLS

Programming	Python, C++, Matlab, R, JavaScript, Java, CSS
Frameworks	PyTorch, NumPy, Detectron2, scikit-image, MMDetection, MONAI, scikit-learn